## ■ 2.2.4 Symbolic Indices

You can think of an expression such as `a[k]` as being like an "indexed variable", where instead of being a number, the index is now the symbol `k`.

This defines a value for `a[k]`.

```
In[1]:= a[k] = 4
Out[1]= 4
```

You can define values for `a[i]` with other "indices", symbolic or numerical.

```
In[2]:= a[0] = a[kp] = 0
Out[2]= 0
```

This shows all the values associated with `a` so far.

```
In[3]:= ?a
a
a/: a[k] = 4
a/: a[kp] = 0
a/: a[0] = 0
```

You can use definitions for objects with symbolic indices to build up simple "databases".

This defines the "object" `area` with "index" `square` to have value 1.

```
In[3]:= area[square] = 1
Out[3]= 1
```

This adds another result to the `area` "database".

```
In[4]:= area[triangle] = 1/2
         1
Out[4]= -
         2
```

Here are the entries in the `area` database so far.

```
In[5]:= ?area
area
area/: area[square] = 1
                          1
area/: area[triangle] = -
                          2
```

You can use these definitions wherever you want. You have not yet assigned a value for `area[pentagon]`.

```
In[5]:= 4 area[square] + area[pentagon]
Out[5]= 4 + area[pentagon]
```

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